

EDULEARN

Student Learning Management Platform

PROJECT REPORT

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1 Student Details

Name	Ayush Mishra
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Sub ject	React JS
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Technology	React JS

2 Project Title

EduLearn – Student Learning Management Platform

3 Problem Statement

Educational platforms require an efficient way to manage course content, track learning progress, conduct quizzes, and maintain student engagement.

Students often face difficulties in tracking completed lessons, resuming videos from their last watched position, reviewing grades, and accessing discussions related to course content.

The EduLearn platform aims to solve these challenges by providing a centralized dashboard where students can access lessons, monitor progress, participate in quizzes, view grades, earn badges, and interact with learning communities.

4 Objectives

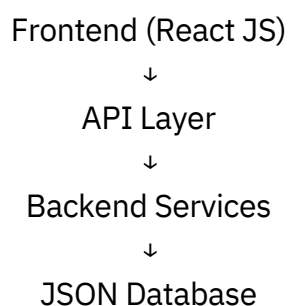
- Create a centralized learning dashboard.
- Track lesson completion progress.
- Save and restore video progress.
- Conduct guided quizzes.
- Display grades efficiently.
- Generate achievement badges.
- Manage student permissions.
- Provide discussion access.
- Support dark and light themes.

5 Tech Stack

Technology	Purpose
React JS	Frontend Development
Vite	Build Tool
Node JS	Runtime Environment
Express JS	Backend Services
JSON Storage	Data Persistence
CSS	Styling
Lucide React	Icons
GitHub	Version Control

6 Component Architecture

EduLearn follows a modern client-server architecture.



The React frontend is responsible for rendering the user interface, lesson dashboards, quizzes, gradebooks, and discussion sections.

The backend processes requests, stores progress information, manages permissions, and handles quiz data.

The database stores lesson information, quiz responses, progress records, grades, and user preferences.

7 Features

7.1 Expandable Lesson List

The lesson list organizes course content into multiple modules. Students can expand and collapse lesson groups while monitoring completion progress through visual indicators.

7.2 VideoProgressMarker

The platform automatically stores the exact video position of a student. Users can leave a lesson and later continue from the same timestamp without losing progress.

7.3 GuidedQuizCreator

The quiz module provides a structured step-by-step question flow. Student responses are stored automatically, ensuring that answers remain available throughout the assessment process.

7.4 Searchable Gradebook

The gradebook allows students to quickly review scores and academic performance. Efficient searching improves accessibility for large course datasets.

7.5 Instant Badge Generator

Students receive digital badges after completing specific learning milestones. This feature encourages engagement and motivates continuous learning.

7.6 Student Access Center

The access control module manages permissions and determines which lessons, quizzes, and resources are available to different users.

7.7 Q&ADiscussionForum

Students can participate in discussions related to course content. This encourages collaboration, problem solving, and peer-to-peer learning.

7.8 Theme Switcher

Users can switch between light mode and dark mode to improve readability and personalize their learning experience.

8 Screenshots of Execution

8.1 DashboardScreen

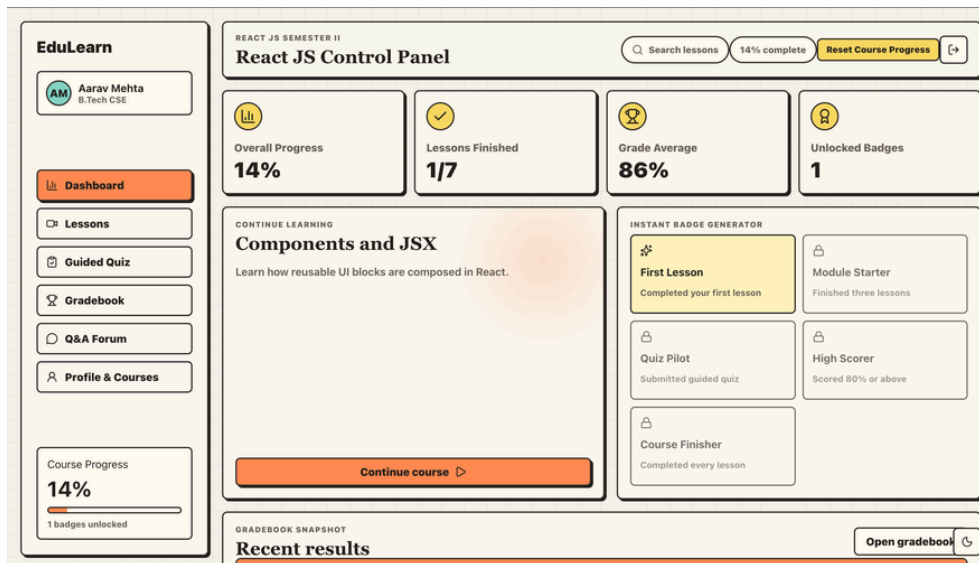


Figure 1: EduLearn Dashboard

8.2 Lesson Progress Screen

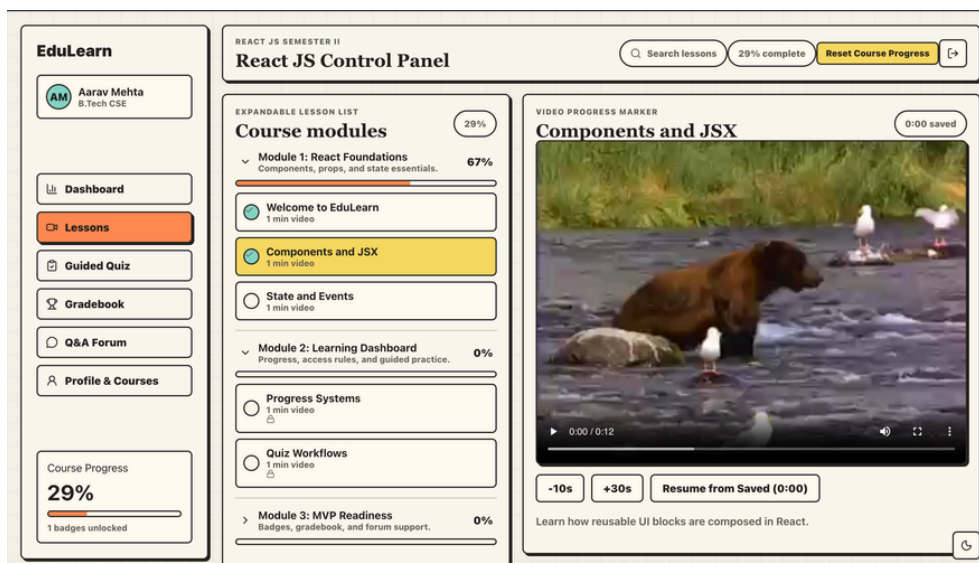


Figure 2: Lesson Progress Tracking

8.3 QuizInterface

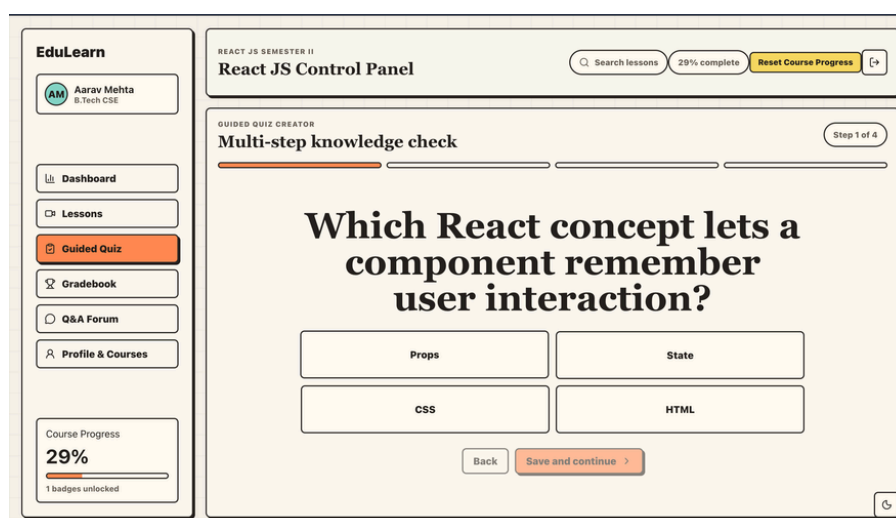


Figure 3: Guided Quiz System

9 Results and Findings

The EduLearn platform successfully demonstrates the implementation of a modern learning management system using React-based web technologies.

The expandable lesson system improves course organization and enables students to track learning progress efficiently.

The video progress marker ensures continuity of learning by preserving playback positions.

The guided quiz creator simplifies assessment management and automatically stores user responses.

The searchable gradebook allows students to access academic results quickly, while the badge generation system increases motivation through achievement recognition.

The discussion forum enhances collaboration and creates an interactive educational environment.

10 Conclusion

EduLearn successfully provides a complete student learning management platform.

The system integrates lesson tracking, video progress management, quizzes, gradebooks, badges, discussion forums, theme customization, and permission control within a single application.

The project demonstrates practical use of modern frontend and backend development technologies while delivering an efficient, responsive, and user-friendly educational experience.

11 GitHubRepositoryLink

<https://github.com/Ayusshh020/EduLearn>

12 Live Demo Link

<https://edu-learn-mocha.vercel.app/>